Spring Semester 2026 EE209 Midterm Exam

Pledging of No Cheating

Note: Please write down your student ID and name, sign on it (draw your signature), and date it. If you do not fill out this page, we won't grade your mid-term exam.

저는 이번 시험을 치르면서 이 과목에서 금지한 어떤 부정행위도 저지르지 않을 것임을 서약합니다. 추후에 위반사항이 발견되었을 경우 합당한 모든 불이익을 감수하겠습니다.

I pledge that I will not participate in any activity of cheating disallowed by this course while taking this exam online. I will assume full responsibility if any violation is found later.

Student ID: _____ Name: _____

Signature: _____ Date: _____

The exam is closed book and notes. Read the questions carefully and focus your answers on what has been asked. You are allowed to ask the instructor/TAs for help only in understanding the questions, in case you find them not completely clear. Be concise and precise in your answers and state clearly any assumption you may have made. All your answers must be included in the attached sheets. You have 150 minutes to complete your exam. Be wise in managing your time.

Please do not fill in the "Score" fields below. Self-grading is not allowed. Good luck!

1	/10
2	/10
3	/10
4	/10
5	/10
6	/10
7	/10
8	/10
9	/10
Total	/90

1. C Fundamentals

① (1 point for each) True or False

i. Overflow in unsigned integers is well-defined in C language.
Your answer: (True / False) True

ii. In two's complement, there are two representations for zero.
Your answer: (True / False) False

iii. In an expression mixing signed and unsigned integers, the signed integer is converted to unsigned.
Your answer: (True / False) True

iv. sizeof(int) returns a signed integer.

Your answer: (True / False) False

② (2 points) What is the output of the following code?

```
#include <stdio.h>

int main() {
    int x = 2;
    switch(x) {
        case 1: printf("A ");
        case 2: printf("B ");
        case 3: printf("C ");
        default: printf("D");
    }

    return 0;
}
```

Your answer: _____ BCD

③ (2 points) What is the output of the following code?

```
#include <stdio.h>

int main() {
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            if (j == 1) break;
            printf("%d%d ", i, j);
        }
    }
    return 0;
}
```

Your answer: _____ 00 10 20

④ (2 points) What is the output of the following code?

```
#include <stdio.h>

int main() {
    int a[] = {1, 2, 3, 4};
    int *p = a;

    printf("%d₩n", *(p + *(p + 2)));
    return 0;
}
```

Your answer: _____ 4

2. Numbers in Computers

① (1 point) Convert decimal 13 to binary.

Your answer: _____ 1101

② (1 point) Convert the binary 10110_2 to decimal.

Your answer: _____ 22

③ (1 point) Convert the binary 11101010_2 to hexadecimal.

Your answer: _____ EA

④ (1 point) Find the 8-bit 2's complement representation of -5 in binary.

Your answer: _____ 11111011

⑤ (1 point) Compute the result of −50 + (−80) in decimal, assuming both numbers are represented as 8-bit signed integers (2's complement).

Your answer: _____ 30

⑥ (1 point) What is the result of 'printf("%d₩n", -10 < 5U);'?

Your answer: _____ 0

⑦ (1 point) Does the following code detect overflow correctly?

```
int a = INT_MAX;
int b = a + 1;
if (b < a) printf("overflow₩n");
```

Your answer: (Yes / No) No

⑧ (1 point) What is the value of a?

```
int a = 0;
int x = 0;
if (x & 1 == 0)
    a = 1;
```

Your answer: _____ 0

⑨ (1 point) What is the output of the following code?

```
unsigned int x = (unsigned int)-1;
int y = (int)x;
printf("%d₩n", y);
```

Your answer: _____ -1

⑩ (1 point) What is the output of the following code?

```
int a = 5, b = 3;
a = a ^ b; b = a ^ b; a = a ^ b;
```

Your answer: a =_____ b =_____

a = 3 b = 5 (1 point only when both a and b are correct, otherwise zero point)

3. C Data Types

① (1 point) Find a bug in the following code and fix it.

```
char c = getchar();
if (c == EOF)
    printf("EOF₩n");
```

Your answer:

char c should be changed to int c

② (1 point) What is the output of the following code?

```
unsigned short u = 65535;
printf("%hu₩n", u); printf("%hd₩n", u);
```

Your answer: _____ 65535 -1

③ (1 point) What is the output of the following code?

```
char s[] = "abcde";
printf("%lu %lu₩n", sizeof(s), strlen(s));
```

Your answer: _____ 6 5

④ (1 point for each) Determine the data type of each constant.

i. 123.456

Your answer: _____ double

ii. 123.456F

Your answer: _____ float

iii. 123.456L

Your answer: _____ long double

iv. 1E-2

Your answer: _____ double

⑤ (1 point) Which statement is True?

A. char is not an integer type
B. Strings are a built-in type in C
C. char can be used in arithmetic
D. sizeof("a") == 1

Your answer: _____ C

⑥ (2 point) Explain the difference between 'a' vs "a".

Your answer:

'a': single character (integer value 97)

"a": string → {'a', '₩0'}

4. Functions

① (2 points) What is the output of f(3)?

```
void f(int n) {
    if (n == 0) return;
    printf("%d ", n);
    f(n - 1);
    printf("%d ", n);
}
```

Your answer: _____ 3 2 1 1 2 3

② (2 points) Fill in the blank with recursion using only f() such that f(5) = 8, f(10) = 89, and f(16) = 1,597.

```
int f(int n) {
    if (n <= 1) return 1;
    return blank;
}
```

Your answer: _____ f(n-1) + f(n-2)

③ (4 points) Fill in the blank_[a] and blank_[b] such that the following program prin ts 10.

```
int f(int *p, int n) {
    if (n == 0) return 0;
    return *p + f(p + 1, blank_[a]);
}

int main() {
    int a[] = {1, 2, 3, blank_[b]};
    printf("%d₩n", f(a, 4));
}

output: 10
```

Your answer: blank_[a]: _____ blank_[b]:_____
n -1, 4 (2 points for each)

④ (2 points) What is the output of f(12)?

```
int f(int n) {
    static int x = 0;
    if (n == 0) return x;
    x += n;
    return f(n - 1);
}
```

Your answer: _____ 78

5. Scope, Array, and String

Consider the following two C files.

```
/* counter.c */
static int count = 0; /* line 1 */
void increment() {
   count++;
}
int get() {
   return count;
}
```

```
/* main.c */
#include <stdio.h>
extern int count;
void increment();
int get();
int main() {
    increment();
    increment();
    printf("%d %d₩n", get(), count);
    return 0;
}
```

① (1 point) Will this program compile and link?

Your answer: _____ No

② (1 point) What will it print??

Your answer: _____

No output (program does not successfully link, so it cannot run)

③ (1 point) What is the output of the program if static were removed from line 1 of counter.c?

Your answer: _____ 2 2

④ (2 points) Briefly explain lexical scope and temporal scope in one sentence for each.

Your answer:

Lexical scope: the region of code where a variable name is visible and can be used. (1 point)

Temporal scope: the period of program execution during which the variable's memory is allocated and its value persists. (1 point)

⑤ (2 points) What is the output of the following code?

```
int x;

void f() {
    int x = x + 1;
    printf("%d₩n", x);
}

int main() {
    f();
    printf("%d₩n", x);
    return 0;
}
```

Your answer: _____ <garbage value> 0

⑥ (3 points) Identify which declaration of i is used at each marked line.

```
int i = 10;              /* Declaration A */
void g(void) {
    int i = 20;          /* Declaration B */
    if (i > 0) {
        int i;          /* Declaration C */
        i = 30;          /* Line X */
    }
    i = 40;              /* Line Y */
}
void h(void) {
    i = 50;              /* Line Z */
}
```

Your answer:

Line X:_____ Line Y:_____ Line Z:_____

Declaration C, Declaration B, Declaration A (1 point for each)

6. Pointer

① (1 point) What is the type of the variable p in 'char (*p)[4];'?

Your answer: _____

p is a pointer to an array of four characters.

② (1 point) Breifly describe what is wrong with the following code.

```
int a[3], b[3];
if (&a[1] > &b[1])
    printf("yes");
```

Your answer:

The code compares pointers from two different arrays, which is undefined behavior.

③ (2 points) What is the output of the following code?

```
int x = 10, y = 20;
int *p = &x;
int **q = &p;

*q = &y;
**q = 30;

printf("%d %d₩n", x, y);
```

Your answer: _____ 10 30 (1 point for each)

④ (2 points) Fill in the blank so that the function reverse() correctly reverses an array of integers in-place.

```
void reverse(int *a, int n) {
    int *p = a;
    int *q = blank;

    while (p < q) {
        int tmp = *p;
        *p = *q;
        *q = tmp;
        p++;
        q--;
    }
}
```

Your answer: _____ a + n – 1 or &a[n - 1]

⑤ (2 points) Complete the two blanks blank_[a] and blank_[b] using pointer p so th at the function count_even() correctly counts even numbers in the array.

```
int count_even(int *a, int n) {
    int count = 0;
    int *p = a;

    for (int i = 0; i < n; i++) {
        if (blank_[a] % 2 == 0)
            count++;
        blank_[b];
    }
    return count;
}
```

Your answer: blank_[a]: _____ blank_[b]:_____

*p or p[0] (1 point)

p++ or ++p or p += 1 or p = p + 1 (1 point)

⑥ (2 point) What is the output of the following code?

```
#include <stdio.h>

void update(int *p, int *q) {
    *p += 2;
    q++;
    *q += *p;
}

void shift(int **pp) {
    (*pp)++;
    **pp += 3;
}
```

```
int main() {
    int A[4] = {1, 2, 3, 4};
    int B[4] = {10, 20, 30, 40};
    int *p = A;
    int *q = B;

    update(p, q);
    shift(&p);
    *(p + 1) += *(q + 2);
    *q += *p;

    printf("%d %d₩n",
        A[0] + A[1] + A[2] + A[3],
        B[0] + B[1] + B[2] + B[3]);

    return 0;
}
```

Your answer: _____ 45 108 (1 point for each)

7. Structures

① (1 points) Fill in the blank to dynamically allocate exactly enough memory for one Student struct.

```
typedef struct {
    int id;
    char *name;
} Student;

Student *s = blank
```

Your answer: _____

(Student *)malloc(sizeof(Student)); or malloc(sizeof(Student));

② (1 point) What is the output of the following code?

```
typedef struct {
    char *name;
} S;

int main() {
    S s1, s2;
    s1.name = "DEF";
    s2 = s1;
    s2.name = "321";
    printf("%s%s₩n", s1.name, s2.name);
}
```

Your answer: _____ DEF321

③ (2 points) Does the following code compile without errors? If not, identify the error and briefly explain its cause.

```
struct A { int x; };
struct B { int x; };
struct A a = {5};
struct B b;
b = a;
```

Your answer:

No (1 point)

Compilation error: incompatible types. Even though fields match, types are different (1 point)

④ (2 points) Identify the bug in the following code and provide a concise explanati on of the fix.

```
char **arr = malloc(3 * sizeof(char *));
for (int i = 0; i < 3; i++)
    arr[i] = malloc(10);

free(arr);
for (int i = 0; i < 3; i++)
    free(arr[i]);
```

Your answer:

Bug: Freeing inner pointers after the outer Array (1 point)

Fix: Free the inner pointers first, then free the outer array: (1 point)

⑤ (4 point) What is the output of the following code?

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

typedef struct {
    char *name;
    int *scores;
} Student;

int main() {
    Student s1, s2;

    s1.name = malloc(10);
    strcpy(s1.name, "John");

    s1.scores = malloc(3 * sizeof(int));
    for (int i = 0; i < 3; i++)
        s1.scores[i] = i + 1;

    s2 = s1;
    s2.name = malloc(10);
    strcpy(s2.name, "Mike");
    s2.scores[0] = 100;

    printf("%s %d %s %d₩n", s1.name, s1.scores[0], s2.name, s2.scores[0]);

    free(s1.name);
    free(s2.name);
    free(s1.scores);
}
```

Your answer: _____

John 100 Mike 100 (1 point for each)

8. Stacks and Queues

① (1 point) What is the output of the following code?

```
#include <stdio.h>

#define STACK_SIZE 5

int contents[STACK_SIZE];
int top = 0;

void push(int v) {
    contents[top++] = v;
}

int pop() {
    return contents[--top];
}
```

```
int main() {
    push(1);
    push(2);
    push(3);
    printf("%d ", pop());
    push(4);
    printf("%d ", pop());
    printf("%d ", pop());
    return 0;
}
```

Your answer: _____ 3 4 2

② (1points) Identify the bug in the following push() function of a stack implementation and briefly explain the issue.

```
void push(int value) {
    contents[top] = value;
    top++;
}
```

Your answer:

Missing overflow check.

③ (1 points) Why is random access not allowed in stack?

Your answer:
Violates LIFO (Last In First Out) abstraction

④ (2 points) Based on the concept of modularity discussed in class, identify and briefly explain the potential modularity issues in the following code.

```
/* stack.h */
#define STACK_SIZE 100

struct Stack {
  char contents[STACK_SIZE];
  int top;
};

struct Stack *Stack_new(void);
void  Stack_free(struct Stack *s);
void  Stack_push(struct Stack *s, const char *item);
char *Stack_top(struct Stack *s);
void  Stack_pop(struct Stack *s);
int   Stack_isEmpty(struct Stack *s);
```

Your answer:

Interface reveals how the stack is implemented (e.g., as an array) (1 point)

Client can access/change data directly; could corrupt the object (1 point)

⑤ (1 points) Consider the following implementation of the enqueue() function. Identify the bug and provide a corrected version of the code.

```
void enqueue(Queue *q, int v) {
    if (q->size == QUEUE_SIZE)
        return;
    q->rear = (q->rear + 1) % QUEUE_SIZE;
    q->contents[q->rear] = v;
    q->size++;
}
```

Your answer:

Wrong order

q->contents[q->rear] = v;

q->rear = (q->rear + 1) % QUEUE_SIZE;

Consider the following implementation of the Stack_push() function.

```
void Stack_push(Stack *s, int value) {
  assert(s != NULL);
  if (is_full(s))
    stack_overflow();
  else
    s->contents[s->top++] = value;
}
```

⑥ (1 point) If s is NULL, which type of error is it? (Programmer / User)

Your answer: _____ Programmer error

⑦ (1 point) If the stack is full, which type of error is it? (Programmer / User)

Your answer: _____ User error

⑧ (2 points) Fill in the blank to implement a function that checks if parentheses ar e balanced using stack.

```
int is_balanced(char *str) {
    Stack s;
    make_empty(&s);

    for (int i = 0; str[i]; i++) {
        if (str[i] == '(')
            push(&s, '(');
        else if (blank) {
            if (is_empty(&s))
                return 0;
            pop(&s);
        }
    }
    return is_empty(&s);
}
```

Your answer: _____

str[i] == ')'

9. Linked Lists

① (1 point) What is the time complexity of searching in a linked list?

Your answer: _____ O(n)

② (1 points) Discuss the strengths of a linked list.

Your answer:

Anything below is 1 point

- Easy to implement and maintain
  (Increasing or decreasing its length can be done instantly)
- Supports dynamic change of its length

③ (1 points) Discuss the weaknesses of a linked list.

Your answer:

Anything below is 1 point

- Does not support random access

  (Need to traverse the whole list to get the i-th node)

- Does not support instant search

  (Need to traverse the whole list to find a key, e.g., "Mantle")

④ (2 points) Fill in the blank to implement a function that reverses a linked list.

```
struct Node* reverse(struct Node *head) {
    struct Node *prev = NULL, *curr = head;
    while (curr) {
        struct Node *next = curr->next;
        blank  = prev;
        prev = curr;
        curr = next;
    }
    return prev;
}
```

Your answer: _____

curr->next

⑤ (2 points) Explain in English (without code) under what condition the following function returns 1.

```
int f(struct Node *head) {
    struct Node *slow = head, *fast = head;

    while (fast && fast->next) {
        slow = slow->next;
        fast = fast->next->next;
        if (slow == fast)
            return 1;
    }
    return 0;
}
```

Your answer:

This function returns 1 when the linked list has a cycle (a loop).

⑥ (1 points) In the Table_free() function below, why does it need nextp?

```
void Table_free(struct Table *t) {
   struct Node *p;
   struct Node *nextp;
   for (p = t->first; p != NULL; p = nextp) {
      nextp = p->next;
      free(p);
   }
   free(t);
}
```

Your answer:

Because after freeing p, we cannot access p->next

⑦ (2 points) Fill in the blank to implement Table_free() function using a while loop.

```
void Table_free(struct Table *t) {
    struct Node *p = t->first;

    while (p != NULL) {
        struct Node *nextp = p->next;
        free(p);
        blank ;
    }
    free(t);
}
```

Your answer: _____

p = nextp